2022-2023 AIAA Design, Build, Fly Proposal

University of Missouri - Columbia

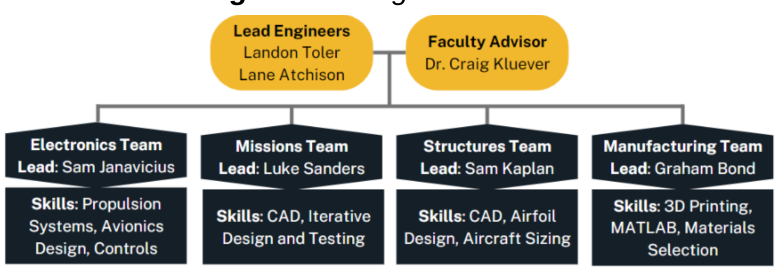
1. Executive Summary

This proposal details the Mizzou AeroTigers' approach to the design, production, testing, and optimization of an unmanned aircraft for the 2022-2023 AIAA Design, Build, Fly Competition. The Mizzou AeroTigers represent the University of Missouri - Columbia College of Engineering. The overall objective is to create an aircraft capable of electronic warfare through modular assembly, surveillance, and jamming. After decomposing mission objectives, a sensitivity analysis was performed to determine the impact of design parameters on the scoring criteria. Various component configurations were assessed, with the highest scoring options developed into a conceptual design. Furthermore, mission-specific components were given detailed designs and preliminary testing to validate their utility. This year's design is a conventional mid-wing single-motor aircraft capable of carrying an electronics package payload of up to **10 pounds** as well as a PVC jamming antenna with a maximum length of **24 inches**. The aircraft will complete a single lap of the nominal flight course in less than **100 seconds**; however, we expect it to complete a lap in as little as **30 seconds** for greater scoring. The following proposal details our organizational structure, timeline, budget, conceptual design, manufacturing plan, and testing plan for the aircraft and its components.

2. Management Summary

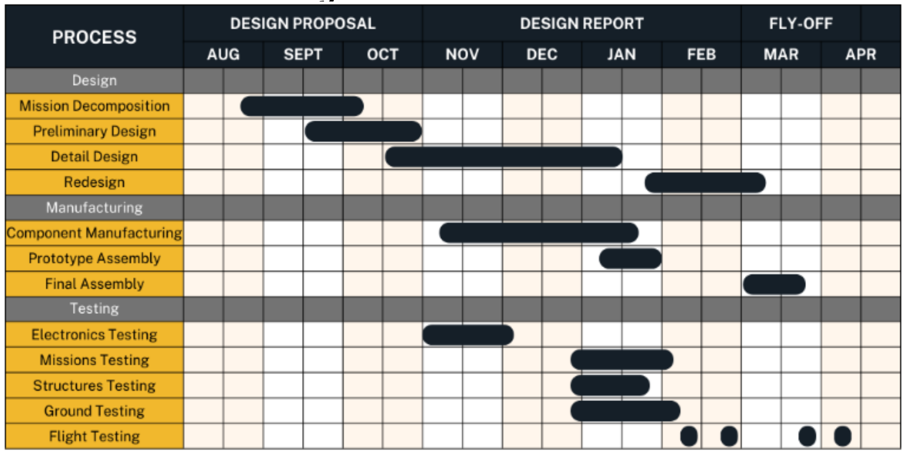
2.1 Team Organization: The Mizzou AeroTigers is an entirely student-led organization separated into four teams: Structures, Electronics, Missions, and Manufacturing. Each team has a student lead responsible for ensuring the team's timely success. The Structures team is responsible for the overall structural design of the aircraft, SolidWorks modeling, planform design, and the testing and validation of aircraft structures. The Electronics team is responsible for the design, testing, and validation of all electronic components including the propulsion, control, and energy storage systems. The Missions team is responsible for setting mission requirements and for the design, testing, and implementation of missions systems. The Manufacturing team is responsible for supporting the final assembly of aircraft subsystems, the ordering of tools and materials, and the simulation of aircraft controls. The teams are organized as shown in **Figure 2.1** and all members are involved in the writing of this proposal and the design report.

Figure 2.1*: Organization Chart*



2.2 Schedule of Major Milestones: Figure 2.2 presents the team's plan to prepare for the fly-off. Maintaining scheduled deadlines is key to executing an effective test plan and producing a high-scoring aircraft in time for the fly-off. The team's mission decomposition and preliminary design are already complete, and section 3 details these components. The detail design phase is next and includes generating detailed sizing and CAD models for each component. Following the manufacturing of a fully functional prototype in January, any components that can be further optimized will undergo a redesign for the final competition aircraft. This manufacturing strategy is further detailed in section 4, while testing procedures and deadlines are contained in section 5.

***Figure 2.2**: Gantt Chart*



2.3 Budget: The Mizzou AeroTigers' anticipated budget for the 2023 competition is detailed in **Figure 2.3**. At this time, more than $3,000 has been received from the College of Engineering, enough to complete the assembly of the aircraft. While little support has been received from the other sources at this time, a combination of fundraising events, corporate outreach, and support from the University's Organizational Resource Group should cover expected travel costs. The planned travel costs allow the team to bring 10 students, consisting of leadership and primary contributors, to the in-person fly-off. However, any extra funding received will allow more team members to attend.

***Figure 2.3**: Budget*

CATEGORY	ITEM	ANTICIPATED COST (USD)
Electronics ($750)	Motors, Propellers	$350
	Control Systems	$200
	Batteries	$200
Structures ($620)	Carbon Fiber Spars	$200
	Balsa Wood, PLA Filament	$350
	Landing Gear, Hardware	$70
Travel ($4,500)	Lodging (4 nights)	$1500
	Transportation	$3000
Additional ($500)	Tools	$400
	Operating Costs	$100
Total Anticipated Cost		**$6,370**
Estimated Support ($3,521)	College of Engineering Support	$3,291
	Corporate Support	$230
	Public Support	$0
Remaining Cost		**$2,849**

3. Conceptual Design

3.1 Decomposition of Mission Requirements: This year's challenge is to develop an aircraft with the ability to execute electronic warfare missions. The aircraft must weigh less than **50 pounds** and disassemble to fit into a shipping box with a maximum length, width, and height sum of **62 inches**. Three flight missions will be conducted to gauge the effectiveness of the aircraft design, and one ground mission will evaluate the structural integrity of the wings and related components. Points awarded for each mission are defined in **Figure 3.1** where $\mathbf{M_n}$ is the individual mission score, **W** is the payload weight, $\mathbf{W_{test}}$ is the marginal ground mission weight, $\mathbf{N_{laps}}$ is the number of laps completed, **L** is the antenna length, and **t** is the mission completion time. To compute a total score, the mission scores will be summed and multiplied by the design report score, which is scored out of 100. A small additional score for participation is also included.

***Figure 3.1**: Mission Decomposition*

MISSION	OBJECTIVE	SCORING	DESIGN PARAMETERS
M1: Staging Flight	• Fit all components into shipping box • Complete 3 laps in 5 minutes	$M_1 = 1.0$	• Ensure unweighted aircraft maintains 5 minute pace
M2: Surveillance Flight	• Complete maximum laps in 10 minutes • Carry maximum payload weight	$M_2 = 1.0 + \frac{[W/N_{laps}]_{MU}}{[W/N_{laps}]_{Max}}$	• Ensure stall speed allows takeoff • Maximize in-flight velocity
M3: Jamming Flight	• Complete 3 laps in minimum time • Carry antenna with maximum length	$M_3 = 2.0 + \frac{[L/t]_{MU}}{[L/t]_{Max}}$	• Balance asymmetric drag profile and CG • Maximize in-flight velocity • Ensure structural integrity of mount
GM: Structural Margin Demonstration	• Maximize testing weight	$GM = \frac{[W_{test}/W_{total}]_{MU}}{[W_{test}/W_{total}]_{Max}}$	• Minimize wingspan to reduce moment • Maximize wing and mounting rigidity

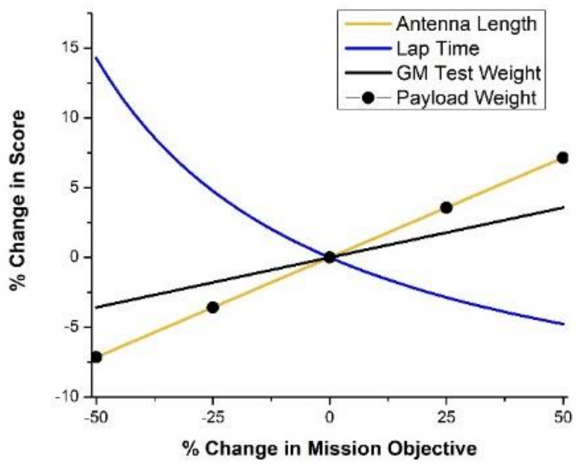


***Figure 3.2**: Score Variation with Mission Outcome*

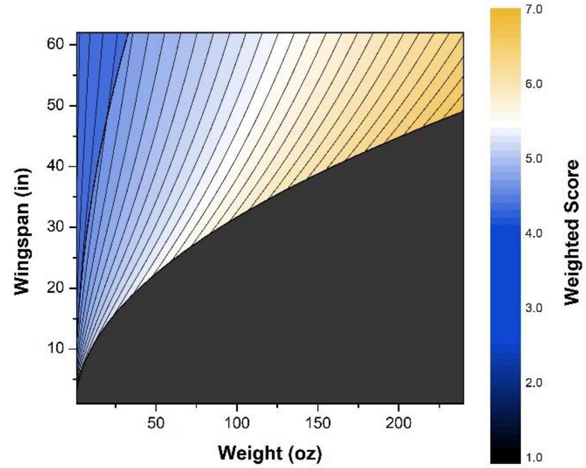


***Figure 3.3**: Score Variation with Design Parameters*

3.2 Sensitivity Study of Design Parameters: To best optimize the aircraft's design, a sensitivity analysis was performed to determine how the design variables impact the scoring criteria. MATLAB was used to perform the analysis, and Origin 8.6 was used to plot the results. **Figure 3.2** depicts how each mission objective impacts the total score, demonstrating that antenna length and payload weight have an equivalent linear relationship, the GM testing weight has a less impactful linear relationship, and lap time has an inverse, nonlinear relationship. Baselines were determined with previous competition outcomes, with the score changes calculated using the scoring equations in **Figure 3.1**. From this analysis, optimizing in-flight velocity to minimize lap time is the most important parameter, with antenna length and payload weight both having an equivalent, secondary impact. With these findings in mind, aircraft weight and wingspan were selected as two of the most important parameters to optimize as they interfered with each other, both impacted in-flight maximum velocity, and both impacted the magnitude of wing loading in mission two and the ground mission. **Figure 3.3** depicts this analysis, with a weighted score given to each possible iteration based on the maximum score of seven points across the four missions. The results indicate that for any given weight a minimum threshold of wingspan must be reached for the in-flight velocity to be fast enough to complete every mission. The most desirable design will thus include the maximum possible weight (and therefore payload weight) and minimum allowable wingspan.

3.3 Preliminary Design: Overarching design configurations were analyzed and compared within **Figure 3.4**. A mid-wing and conventional empennage were selected to target stability, control, and a form factor that is useful for mounting the wings. Due to the shipping box size limitation and ease of assembly required for the ground mission, simplicity in the assembly of aircraft components has been prioritized by selecting a single motor in tractor orientation that will be mounted to the front of the aircraft fuselage. Furthermore, tricycle-style landing gear was selected due to its stability and ease of attaching it to the fuselage. Both the landing gear and wings will be mounted to the fuselage via a slide-on rail system.

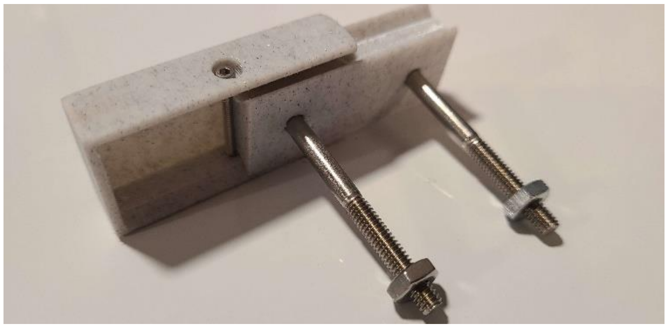
***Figure 3.4**: Configuration Analysis*

COMPONENT	DESIGN CHOICES			CONSIDERATIONS
Wing Location	Low	Middle	High	Stability, Maneuverability, Structural rigidity
Tail Configuration	Conventional	H-Tail	T-Tail	Stability, Control, Strength
Number of Motors	1	2	3 or more	Thrust-to-weight ratio, Integration
Landing Gear Configuration	Conventional	Tricycle	Tandem	Ground stability, Integration
Antenna Mounting Hardware	Hose Clamp	Seat Post Clamp	Set Screws	Strength, Speed of assembly
Wing Attachment Hardware	Fastened	Slotted	Rail Mounted	Strength, Speed of assembly, Uniformity

Through preliminary experimentation with mounting, this style of rail system was determined to be the strongest and fastest to assemble. An initial design of this rail system, with both the female and male attachments, is shown in **Figure 3.5**.

A model of the selected aircraft configuration and preliminary sizing is depicted in **Figure 3.6**. From the sensitivity study, maximum takeoff weight of the aircraft is estimated to be **15 pounds**. Due to wing-loading requirements and limitations on maximum thrust, a wingspan of **60 inches** and a wing chord of **10 inches** were selected. The Selig S8037 airfoil was chosen due to its low-speed lift characteristics, low stall speed, and thickness, allowing for a carbon fiber spar to be installed inside of the wings.

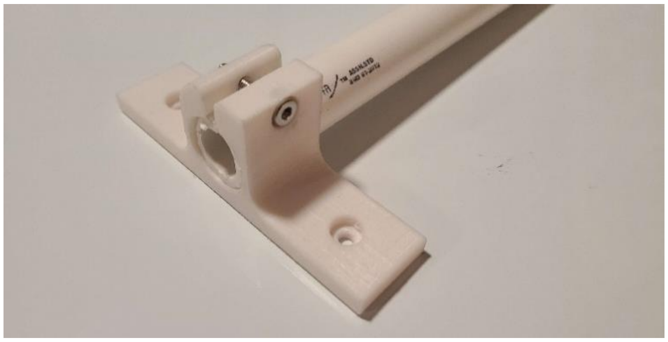
Figure 3.5: *Modular rail mount*



The motor will be center mounted on the fuselage to minimize any propwash effects on the frame. The critical parameters to optimize within the propulsion system are static thrust and battery life, as both are limited by electronics rules. Maximizing static thrust will allow for the plane to easily meet the **60-foot** takeoff constraint while maximizing battery life allows ample time to complete missions with the highest possible score. An initial motor and battery that generates a suitable amount of thrust have been chosen: the Scorpion SII-4020-630 with a 12x6 propeller and a ZIPPY Compact 4000 mAh 6S 25C LiPo battery. This combination complies with the **100-Watt hour** and the **100-amp** discharge restrictions and will be validated to achieve the thrust and battery life requirements through static thrust testing.

Figure 3.6: *Conceptual Design*

Figure 3.7: *PVC Antenna Mount*



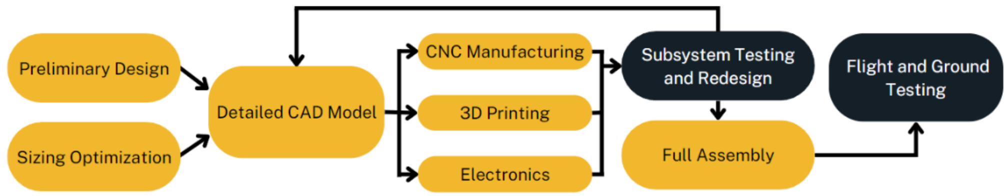
For mission three, multiple antenna-mounting hardware configurations were designed and tested using additive manufacturing. The selected antenna mount closely resembles a bicycle seat post clamp and is shown in **Figure 3.7.** This design achieves the strength and speed of assembly requirements for a 24-inch-long section of 1/2-inch, schedule 40 PVC pipe. With the antenna being installed on only one wing of the aircraft, the difference in drag between the wings will cause a rotation. To counteract this rotation, the drag caused by a 24-inch-long section of antenna has been studied and a lightweight counterbalance that mimics the drag profile of the antenna will be installed on the opposing wing.

4. Manufacturing Plan

The manufacturing flow is outlined in **Figure 4.1**. The primary parameters to optimize within model aircraft manufacturing are weight, mechanical strength, and precision. Since the fly-off requires two wing pairs for each mission, the manufacturing consistency of each wing section is critical. With these objectives in mind, each component will be modeled in SolidWorks and broken down into sections for manufacturing using either a 60W CO2 laser cutter or a 3D printer.

Core structural components, such as the ribs, formers, and the sides of the fuselage, will be laser cut from 1/8-inch plywood to form an overall frame. High-strength carbon fiber rods will be used for the tail boom and to reinforce each wing. Connections between the modular aircraft subassemblies will be 3D printed using PLA, with the infill optimized to maintain strength at a low weight. These connections include the rail systems for wings and landing gear and the jamming antenna mount. Once each subsystem is manufactured, the prototype will be assembled, with MonoKote used as an exterior skin for reducing drag in flight. The final competition aircraft will be assembled in the same way following adjustments made to the initial design.

***Figure 4.1**: Manufacturing Flow*



5. Testing Plan

Prior to full aircraft assembly, each team will subject its designs to thorough testing and validation. The electronic subassembly will be tested on a test bench and the propulsion system will be evaluated on a test stand for optimizing thrust and battery discharge. Each of the 3D printed mounts will have weight continuously added until failure to optimize their design for infill and weight. Once the first full aircraft prototype assembly is complete, each structure will be evaluated based on strength, ease of assembly, and manufacturing consistency. Technical inspection requirements will be emphasized, including the wing tip test and taxiing capabilities. The purpose, methodology, and due date for each test are detailed in **Figure 5.1**.

After thorough testing of the initial prototype, the team will redesign the aircraft based on data collected and qualitative feedback from the pilot. The final aircraft will then be subjected to the same ground and flight tests to prepare for competition. Preliminary flight dates for the prototype aircraft are set for **2/11/2023** and **2/25/2023**, with flight dates for the competition aircraft set for **3/25/2023** and **4/8/2023**. Each flight date will consist of the pilot becoming familiar with the aircraft, fine-tuning of the control systems, and timed trials of the three flight missions to optimize scoring.

***Figure 5.1**: Testing Plan*

TEAM	TEST	METHOD	GOAL	DUE DATE
Electronics	Static Thrust Test	Measure thrust, battery usage, and power on a test stand	Determine most efficient propeller and motor to use	12/5/22
	Electronics Bench Test	Attach all electronic systems on test bench	Achieve a working electronics subsystem before assembly on aircraft	11/21/22
Missions	Mount Loading Test	Continuously add weight to each 3D printed mount until failure	Determine if 3D printing parameters are sufficient for each connection within aircraft	10/10/22
	Assembly Test	Assemble and disassemble aircraft from shipping container	Ensure speed, ease, and consistency of assembly	2/6/23
	Ground Loading Test	Perform Ground Mission	Determine maximum weight margin	2/6/23
Structures	Landing Gear Test	Drop loaded aircraft onto landing gear from progressively increasing heights	Ensure reliability of landing gear	1/23/23
	Wing Tip Test	Hold fully loaded aircraft by wing tips	Ensure structural rigidity of wings	1/23/23
Manufacturing	Aerodynamic Analysis	Analyze aerodynamics in MATLAB	Establish suitable aerodynamic properties	11/14/22
	Wing Variation Analysis	Precisely measure each wing for variation	Ensure manufacturing consistency of all four wings	2/6/23
All Teams	Taxi Test	Test movement of aircraft on ground	Ensure ease of movement and control functions	2/11/23
	Flight Test	Pilot aircraft around mock competition course	Practice flight missions, maximize in-flight velocity, and ensure reliability of controls, landings, and batteries	2/11/23 2/25/23 3/25/23 4/8/23
	Mock Technical Inspection	Perform thorough technical inspections of every aspect of the aircraft's construction	Demonstrate ability to pass technical inspection at competition	3/20/23